

Innovation-compressed common feature, $SU(2)$ fibre/covariance split, and signed collar band

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1. Purpose and scope

R-141 reduced the controlled-shell programme to the actual common-heat paired-replica/global-Doob signed Hessian, with the R-102 product covariance, primitive trace, R-125 variance/forest coordinate, returned mean/low owner, and coherent $C = 8$ or $C = 10$ band kept together. The present checkpoint opens that matrix rather than adding another scalar absolute-value budget.

There are four durable advances.

1. The complete mixed-endpoint signature (U, J, R) compresses exactly to two features (U, Φ) , where $\Phi = \mathbb{E}_f J$. The R-102 product covariance is retained inside $D\Phi$ and $D^2\Phi$.
2. The actual covariance/trace feature U is pinned by a shared Gaussian trace probe, including singular covariance factors. This repairs the equal-norm-lift information gap left open in R-131 and R-141.
3. The six-row A1 coefficient has an exact $SU(2)$ fibre block, while the registered A1 family-lock Gaussian block has a scalar principal covariance plus a positive anisotropic remainder four spatial powers smoother at high frequency.
4. A coefficient-exact legal scalar adapted chart is uniformly positive, but the exact moved $C = 8/C = 10$ rational collar band has a negative finite-cutoff direction. These facts isolate, rather than decide, the missing cross-channel production cancellation.

All statements are at positive floor and finite cutoff. The trace feature and Hessian identities are fixed-chart owner identities. The scalar chart is a coefficient-exact active scalar packet, not the full A1 heat law. The negative collar direction is an A1 coefficient-exact deterministic scalar-slice subblock, not a counterexample to the complete owner. No cutoff/chart/control/visit/refinement-uniform upper Loewner bound, source-null and low-kernel production estimate, positive graph gap, matching, absolute anchor, A13 gate, Nelson theorem, removal, interacting measure, Sector-A closure, or tier promotion is claimed.

2. Exact mixed-endpoint innovation compression

Fix one completely recombined owner and one common future conditional projection $P_f = \mathbb{E}[\cdot \mid \mathcal{F}_f]$. At every endpoint q , put

$$\Phi_q = P_f J_q, \quad R_q = (I - P_f) J_q. \quad (2.1)$$

For arbitrary endpoints q, p , orthogonality to the same conditional subspace gives

$$\mathbb{E}_f \langle J_q, J_p \rangle = \langle \Phi_q, \Phi_p \rangle + \mathbb{E}_f \langle R_q, R_p \rangle. \quad (2.2)$$

Consequently,

$$\boxed{\mathbb{E}_f\{-\langle J_q, J_p \rangle + \langle R_q, R_p \rangle\} = -\langle \Phi_q, \Phi_p \rangle.} \quad (2.3)$$

Thus the R-141 signature

$$(U, J, R), \quad S_3 = \text{diag}(+I, -I, +I)$$

has the exact compressed mixed kernel

$$K(q, p) = \mathbb{E}_f\langle U_q, U_p \rangle - \langle \Phi_q, \Phi_p \rangle = \mathbb{E}_f\langle F_q, S_2 F_p \rangle, \quad (2.4)$$

where

$$F_q = (U_q, \Phi_q), \quad S_2 = \text{diag}(+I, -I).$$

This is a recombination, not a deletion. In particular, the innovation product identified in R-102 becomes part of the derivatives of Φ . Using different projections at q and p , or applying (2.3) pointwise before future expectation, is invalid.

2.1 Canonical actual trace feature

Let an owner be indexed by the derivative direction i , output cluster $\mathcal{C}$, heat coordinate r , and retained future f . Let Γ_o be its positive semidefinite covariance factor, possibly singular. On an auxiliary standard Gaussian probe ζ_o , define

$$G'_o = \Gamma_o^{1/2} \zeta_o$$

and use the *same* heat, owner, output space, future coupling, and probe at both endpoints:

$$U_{q,o} = P_{\mathcal{C}} C_6(W_o^{f,r}(q)) G'_o. \quad (2.5)$$

Then

$$\begin{aligned} \mathbb{E}_{\zeta} \|U_q\|^2 &= \sum_o \mathbb{E} \text{Tr}[P_{\mathcal{C}} C_6(W_q) \Gamma_o C_6(W_q)^* P_{\mathcal{C}}] = \Theta_q, \\ \mathbb{E}_{\zeta} \langle U_q, U_p \rangle &= \sum_o \mathbb{E} \text{Tr}[P_{\mathcal{C}} C_6(W_q) \Gamma_o C_6(W_p)^* P_{\mathcal{C}}]. \end{aligned} \quad (2.6)$$

Equation (2.6) pins the mixed trace correlation. Independent probes at the two endpoints would erase it; an arbitrary lift having only the correct diagonal norm would leave it undetermined.

2.2 Exact complete Hessian

For the fixed-chart complete owner,

$$\mathcal{P}_{\text{comp}}(h) = \frac{1}{2} \|\Phi(h)\|^2 - \frac{1}{2} \|U(h)\|^2. \quad (2.7)$$

Its exact bilinear Hessian is

$$\boxed{Q_{\text{comp}}[z, w] = \langle D\Phi[z], D\Phi[w] \rangle + \langle \Phi, D^2\Phi[z, w] \rangle - \langle DU[z], DU[w] \rangle - \langle U, D^2U[z, w] \rangle.} \quad (2.8)$$

The product-covariance square $D\Phi^* D\Phi$ is favorable in the action. The adverse pieces are the trace square and connection together with the two second-jet connection terms; none may be dropped. Formula (2.8) is the R-132 paired-replica Hessian written in the compressed current coordinate.

Both U and Φ factor through the physical endpoint. Hence every physical first or second derivative with one argument in $\text{Ker } L_{\pi}$ vanishes. The explicit source Hessian 9I/10 still fills that kernel. This proves physical source-null factorisation of the compressed owner, but not the low-coordinate compatibility or a uniform quotient bound.

3. The first legal finite matrix

The smallest nontrivial adapted chart uses independent standard Gaussians ξ_1, ξ_2 and two visits,

$$\begin{aligned} h_1 &= a, & h_2(\xi_1) &= b + c\xi_1, \\ Z_\theta &= S_1(\xi_1 + a) + S_2(\xi_2 + b + c\xi_1), & \theta &= (a, b, c). \end{aligned} \quad (3.1)$$

The source basis $(1, 0), (0, 1), (0, \xi_1)$ is orthonormal. The physical tangents are

$$z_1 = S_1, \quad z_2 = S_2, \quad z_3 = \xi_1 S_2. \quad (3.2)$$

Let $F = (U, \Phi)$ be the direct sum over all owners. The complete owner block is

$$C_{pq} = -\langle F_p, S_2 F_q \rangle - \langle F, S_2 F_{pq} \rangle, \quad (3.3)$$

while the terminal sextic block is

$$S_{pq} = \frac{9}{10} \mathbb{E} \int \left\{ |Z|^4 z_p \cdot z_q + 4|Z|^2 (Z \cdot z_p)(Z \cdot z_q) \right\}. \quad (3.4)$$

Therefore the exact finite-chart action matrix is

$$\boxed{H_{pq} = \frac{9}{10} \delta_{pq} + C_{pq} + S_{pq}.} \quad (3.5)$$

The global source-Doob coordinate is already diagonal in (3.1), so the source metric is I_3 . By contrast, the physical synthesis metric is

$$L_\pi^* L_\pi = \begin{pmatrix} \langle S_1, S_1 \rangle & \langle S_1, S_2 \rangle & 0 \\ \langle S_2, S_1 \rangle & \langle S_2, S_2 \rangle & 0 \\ 0 & 0 & \langle S_2, S_2 \rangle \end{pmatrix}. \quad (3.6)$$

This distinction explains why a physical synthesis kernel still receives the explicit 9/10 source cost.

Equations (3.1)–(3.6) are a complete assembly recipe, not a production numerical spectrum. The present repository authorities do not yet contain all production U, Φ jets, cross-Grams, trace partitions, Fourier blocks, output projectors, adapted feedback, and returned low blocks needed to fill every entry. Inventing those entries would be a false numerical closure.

4. Exact $SU(2)$ fibre block

Use the production Class-II coefficients

$$P = 4 + 10^{-12}, \quad \alpha = \frac{5}{9}, \quad c_0 = \frac{3}{250P}, \quad c_1 = \frac{243}{8000P}. \quad (4.1)$$

Write $z = (\psi, \chi) \in \mathbb{C}^2 \oplus \mathbb{C}$. Rotate one fibre endpoint to $\psi = (r, 0)$, choose $\chi = s \geq 0$, and set

$$d = r^2 + s^2 + e, \quad t = \frac{r^2}{d}. \quad (4.2)$$

In the active tangent coordinates (x_1, x_2, x_r, x_s) , the full six-row coefficient map is

$$C_0 = 2 \begin{pmatrix} \sqrt{c_0}r & 0 & 0 & 0 \\ 0 & \sqrt{c_0}r & 0 & 0 \\ 0 & 0 & \sqrt{c_0}r & 0 \\ \sqrt{c_1}r & 0 & 0 & 0 \\ 0 & \sqrt{c_1}r & 0 & 0 \\ 0 & 0 & \sqrt{c_1}r(1 - \alpha t) & -\sqrt{c_1}\alpha t s \end{pmatrix}. \quad (4.3)$$

The doublet and singlet phase directions are exact zero columns. The Gram $B = C_0^* C_0$ has two transverse eigenvalues

$$\lambda_T = 4(c_0 + c_1)r^2 = \frac{339}{2000P}r^2 \quad (4.4)$$

and radial block

$$B_{\text{rad}} = 4 \begin{pmatrix} r^2[c_0 + c_1(1 - \alpha t)^2] & -c_1 \alpha t(1 - \alpha t)rs \\ -c_1 \alpha t(1 - \alpha t)rs & c_1 \alpha^2 t^2 s^2 \end{pmatrix}. \quad (4.5)$$

If the inner entries of (4.5) are A, C, D , its eigenvalues and determinant are

$$\lambda_{\pm} = 2\{A + D \pm [(A - D)^2 + 4C^2]^{1/2}\}, \quad (4.6)$$

$$\det B_{\text{rad}} = 16c_0 c_1 \alpha^2 t^2 r^2 s^2. \quad (4.7)$$

Thus the generic rank is four, with two exact phase kernels and additional degeneracy at $r = 0$ or $s = 0$. Moreover,

$$0 \preceq B(z) \preceq 4(c_0 + c_1)r^2 \mathbf{I} \preceq \frac{339}{2000P} \rho \mathbf{I}. \quad (4.8)$$

Equality is attained by either transverse direction. There is no uniform positive lower bound.

The geometric content of (4.3) is a weighted S^2 sigma model plus a two-dimensional radial metric. A moving Bloch frame introduces a skew Cartan/Berry connection; it reorganises the self-adjoint Jacobi operator but does not remove it. Pairwise endpoint gauge reduction leaves one spectator channel and an explicit 3-by-3 block. A generic graph of three or more reveals cannot use one such gauge on every edge unless all Bloch directions lie on one great circle.

This also identifies a strict scope firewall. On the scalar ray $\psi = (Z, 0), \chi = 0$ and $t \rightarrow 1$, the ratio between a transverse coefficient and the tested radial coefficient is

$$\frac{c_0 + c_1}{c_0 + c_1(1 - \alpha)^2} = \frac{339}{144} = \frac{113}{48} > 2.35. \quad (4.9)$$

Consequently, positivity of the scalar chart below cannot certify the transverse or radial-singlet production blocks.

5. Registered A1 covariance: scalar sea plus smoother anisotropy

The registered A1 production parameter set has family masses $(0, 3/100, 7/100)$, lock $3/20$, and normalized lock vector $(1, 1, 1)$. The corresponding internal Gaussian mass block is therefore

$$M = \begin{pmatrix} 1/10 & -1/20 & -1/20 \\ -1/20 & 13/100 & -1/20 \\ -1/20 & -1/20 & 17/100 \end{pmatrix}. \quad (5.1)$$

Its characteristic polynomial is

$$\frac{25000\mu^3 - 10000\mu^2 + 1115\mu - 24}{25000}, \quad (5.2)$$

with eigenvalues

$$\begin{aligned} \mu_1 &= 0.0281161723351\dots, \\ \mu_2 &= 0.165197152881\dots, \\ \mu_3 &= 0.206686674784\dots \end{aligned} \quad (5.3)$$

For $T_3 = \text{diag}(1, -1, 0)$,

$$[M, T_3] = \begin{pmatrix} 0 & 1/10 & 1/20 \\ -1/10 & 0 & -1/20 \\ -1/20 & 1/20 & 0 \end{pmatrix}, \quad \|[M, T_3]\|_{\text{op}}^2 = \frac{3}{200}. \quad (5.4)$$

The stationary production covariance is therefore not $SU(2)$ invariant.

For every scalar kinetic eigenvalue $a_k \geq 0$, however, there is an exact positive split

$$C(k) = (a_k \mathbf{I} + M)^{-1} = \gamma_0(k) \mathbf{I} + R(k), \quad \gamma_0(k) = \frac{1}{a_k + \mu_3}, \quad R(k) \succeq 0, \quad (5.5)$$

with sharp norm

$$\|R(k)\| = \frac{\mu_3 - \mu_1}{(a_k + \mu_1)(a_k + \mu_3)}. \quad (5.6)$$

When the already registered fourth-order high-frequency premise $a_k \asymp |k|^4$ is used,

$$k_i^2 \gamma_0(k) = O(|k|^{-2}), \quad k_i^2 R(k) = O(|k|^{-6}). \quad (5.7)$$

Thus the symmetry-breaking derivative covariance is four powers smoother than the scalar divergent sea. This selects a concrete successor: keep the scalar principal sea in the exact common feature, estimate the positive anisotropic high-shell remainder with the stronger collar, and retain its finite low modes in the exact low graph. Equation (5.7) is not yet an owner-preserving intertwiner and does not prove a $\gamma = 4$ production estimate.

6. Coefficient-exact legal scalar adapted chart

Set $S_1 = S_2 = 1/\sqrt{2}$ in the active R-132 scalar packet and use zero common heat. Then

$$Z_c = \frac{(1+c)\xi_1 + \xi_2 + a + b}{\sqrt{2}}, \quad \mu = \frac{a+b}{\sqrt{2}}, \quad v(c) = 1 + c + \frac{c^2}{2}. \quad (6.1)$$

For $e = \delta^2$, define

$$f_\delta(x) = x - \frac{5}{9} \frac{x^3}{x^2 + \delta^2}, \quad |f_\delta(x)| \leq |x|. \quad (6.2)$$

The two active six-row outputs are

$$C_6(xe_1)e_1 = (2\sqrt{c_0}x, 2\sqrt{c_1}f_\delta(x)). \quad (6.3)$$

The matching trace owner gives exactly

$$\mathcal{P}_{\text{comp}} = \frac{1}{2} \|\Phi\|^2 - \frac{1}{2} \Theta = -\frac{1}{2} \text{Var}(C_6(Z_c)e_1). \quad (6.4)$$

Hence the restricted action is

$$\mathcal{A}_\delta = \frac{9}{20}(a^2 + b^2 + c^2) - 2c_0v(c) - 2c_1Q_\delta(\mu, v(c)) \frac{3}{20} \mathbb{E}Z_c^6, \quad (6.5)$$

where

$$Q_\delta(\mu, v) = \text{Var} f_\delta(\mu + \sqrt{v} X).$$

At $\mu = 0, c = 0$, parity kills the translation-covariance cross entries. The Hessian is

$$H_\delta = \begin{pmatrix} 9/10 + \kappa_\delta/2 & \kappa_\delta/2 & 0 \\ \kappa_\delta/2 & 9/10 + \kappa_\delta/2 & 0 \\ 0 & 0 & \lambda_c \end{pmatrix}. \quad (6.6)$$

The translation eigenvectors are $(1, -1, 0)/\sqrt{2}$ and $(1, 1, 0)/\sqrt{2}$. Their eigenvalues are $9/10$ and $9/10 + \kappa_\delta$, with the floor-uniform R-132 bound

$$\frac{9}{10} + \kappa_\delta \geq m_G = \frac{9}{10} - 4c_1(3 + \sqrt{2}) - \frac{[4c_1(2 + \sqrt{2})]^2}{18} = 0.7653207581934908 \dots > \frac{3}{4}. \quad (6.7)$$

For $q(v) = \mathbb{E}f_\delta(\sqrt{v}X)^2$, the normalized variance scores at $v = 1$ give

$$q'(1) + q''(1) = \frac{1}{4}\mathbb{E}[(X^4 - 4X^2 + 1)f_\delta(X)^2] \leq 4. \quad (6.8)$$

Here the second score is $(\partial_v^2 p_v)/p_v$, not $\partial_v^2 \log p_v$. Using $v'(0) = v''(0) = 1$ and $\mathbb{E}X^2 = 1, \mathbb{E}X^6 = 15$, (6.8) yields

$$\begin{aligned} \lambda_c &= \frac{9}{10} + \frac{81}{4} - 2c_0 - 2c_1\{q'(1) + q''(1)\} \\ &\geq \frac{9}{10} + \frac{81}{4} - 2c_0 - 8c_1 \\ &= \frac{1686660000000423}{80000000000020} = 21.083250000000017 \dots \end{aligned} \quad (6.9)$$

Thus the simplest affine covariance feedback is not the obstruction on this legal scalar chart. The parity decoupling in (6.6) need not survive a nonzero translation or general adapted feedback, and (4.9) prevents any extension from this scalar packet to the full $SU(2)$ production field.

7. Exact signed $C = 8/C = 10$ coefficient band

The R-133 one-real Pauli slice takes

$$s = b \cos(Nx), \quad p = \sin(Nx), \quad f_e(s) = s - \frac{5}{9} \frac{s^3}{s^2 + e}. \quad (7.1)$$

After removing the polynomial and constant principal pieces, the exact rational coefficient is, for $n \geq 2$,

$$\begin{aligned} a_n &= 4c_1 e(-1)^{n+1} e^{-2n\kappa} \left\{ \frac{5\delta}{27c} + \frac{25\delta^2}{81c^2} [n + \coth(2\kappa)] \right\}, \\ \delta &= \frac{\sqrt{e}}{b}, \quad \kappa = \operatorname{arsinh} \delta, \quad c = \sqrt{1 + \delta^2}. \end{aligned} \quad (7.2)$$

The coefficient after multiplication by p^2 is

$$g_n = \frac{a_n}{2} - \frac{a_{n-1} + a_{n+1}}{4}, \quad (-1)^{n+1} g_n > 0 \quad (n \geq 3). \quad (7.3)$$

Indeed, if $a_n = (-1)^{n+1} A_n$ with $A_n > 0$, then

$$(-1)^{n+1} g_n = \frac{A_n}{2} + \frac{A_{n-1} + A_{n+1}}{4} > 0. \quad (7.4)$$

The physical output mode is $2nN$. For shell gap $q = m - j$,

$$2^{q-2} < n \leq 2^{q-1}. \quad (7.5)$$

Therefore

q	5	6	7	8	9
n	9:16	17:32	33:64	65:128	129:256.

(7.6)

Thus $C = 8$ moves $n = 9, \dots, 64$, while $C = 10$ moves $n = 9, \dots, 256$. With any positive endpoint-fixed collar weights w_q ,

$$G_{B,C}(y) = 2 \sum_{q=5}^{C-3} w_q \sum_{n \in I_q} g_n \cos(2ny). \quad (7.7)$$

At $y = \pi/2$, $\cos(2ny) = (-1)^n$, and (7.4) gives

$$G_{B,C}(\pi/2) = -2 \sum_{q=5}^{C-3} w_q \sum_{n \in I_q} |g_n| < 0, \quad C = 8, 10. \quad (7.8)$$

Fejer-localized finite Fourier tangents turn (7.8) into a finite-cutoff negative quadratic direction for this coefficient-exact scalar-slice band. The three rational layers can also align at one output:

$$2114970 = 17(124410) = 33(64090) = 65(32538). \quad (7.9)$$

Hence a quadrature or independent-layer shortcut is false even for the actual rational row.

This negative direction is not a full-owner counterexample. It omits the common-future replica, the favorable R-102 product covariance inside $D\Phi$, the canonical trace probe, R-125 variance/forest recombination, returned mean/low, adapted feedback, and all cross-reveal blocks. The complete $C = 8$ or $C = 10$ owner is non-Toeplitz and its sign remains undetermined.

8. Evidence map and route decision

Route	Verified finding	Boundary / decision
Three-feature owner	(U, J, R) compresses exactly to (U, Φ)	Keep $D\Phi, D^2\Phi$; do not re-add raw-current covariance.
Equal-norm trace lift	Insufficient for mixed endpoints	Use the shared covariance square-root probe (2.5).
$SU(2)$ fibre reduction	Exact $2 + 2$ active block and two phase kernels	Berry/Cartan holonomy prevents a global pairwise gauge.
Stationary symmetry	Broken by the registered family-lock mass	Split scalar principal covariance from positive smoother anisotropy.
Scalar adapted feedback	Minimum at least $m_G > 3/4$ and $\lambda_c > 21.08$	Reject it as the simplest obstruction; do not extrapolate transversely.
Moved collar band	Exact negative $C = 8/C = 10$ scalar-slice direction	Full owner still undecided; assemble signed cross blocks.

The decisive next object is no longer an unspecified large Gram. For the scalar-principal covariance, assemble the four active fibre channels of (4.3) with the common features (U, Φ) , all cross reveals, and the returned low block. Add the smoother anisotropic covariance remainder as a separate positive feature, retaining its finite low modes exactly. On the first three moved layers $q = 5, 6, 7$, compute the non-Toeplitz generalized eigenvalue matrix

$$(R_{m,q}^\pm)^* S_2 R_{m,q'}^\mp, \quad q, q' \in \{5, 6, 7\}, \quad (8.1)$$

including the X_z, X_{zw} and U_z, U_{zw} connections. A positive source-quotient edge advances the route; a negative edge returns an owner-resolved adverse eigenvector. This is falsifiable and does not assume the desired inequality.

9. Devil’s-advocate audit

1. **Objection:** Compression (2.3) deletes the load-bearing R-102 product covariance.
DISMISSED. The covariance is moved into the favorable $D\Phi^*D\Phi$ term and the Φ second jet in (2.8). Both primary and independent certificates retain the full Hessian connection.
2. **Objection:** A vector with norm $\Theta^{1/2}$ already defines the trace feature.
UPHELD against that shortcut. It does not define mixed endpoint correlations. The shared probe (2.5)–(2.6) is required and is checked on a singular covariance.
3. **Objection:** The $SU(2)$ rotation proves a global scalar reduction.
UPHELD. The registered family-lock mass has the nonzero commutator (5.4), multi-reveal Berry holonomy remains, and the transverse coefficient can exceed the scalar radial one by 113/48.
4. **Objection:** The large value in (6.9) closes the production Hessian.
UPHELD. It certifies only the centered coefficient-exact active scalar two-root/two-visit chart. Transverse, singlet-radial, translated, adapted-feedback, low, and cross-reveal blocks are absent.
5. **Objection:** The negative symbol (7.8) refutes T-050.
UPHELD. It is a deterministic coefficient/Hessian subblock. The complete signed owner contains favorable covariance and trace/forest/low cross terms that are not present in the slice.
6. **Objection:** The covariance split (5.5) supplies the missing owner intertwiner automatically.
UPHELD. It is an exact stationary spectral split. Its propagation through heat, output projection, owner partition, adaptation, and refinement is the next theorem, not a consequence of spectral calculus alone.
7. **Objection:** The production A1 parameter source is already a full physical closure.
VALID with mitigation. Only the registered discrete parameter and family-lock block are used. The source manifest’s own external-audit and continuum limitations remain unchanged.

10. Result footer

Result ID: A13-CLASSII-INNOVATION-COMPRESSED-COMMON-FEATURE-SU2-COVARIANCE-SIGNED-COLLAR-BAND-BOUNDARY.
Ledger ID: R-142.
Precise statement: exact innovation compression (2.3)–(2.4), actual shared-probe trace feature (2.5)–(2.6), complete two-feature Hessian (2.8), legal finite matrix (3.5), $SU(2)$ fibre block (4.3)–(4.8), registered A1 covariance split (5.5)–(5.7), scalar-chart bounds (6.7)–(6.9), and negative C8/C10 scalar-slice symbols (7.8).
Scope: finite cutoff, positive floor, fixed temporally faithful chart; coefficient-exact active scalar packet and deterministic rational scalar-slice band only where stated.
Dependencies: A1-PRODUCTION-FUNCTIONAL-REALISATION; R-102, R-125, R-131, R-132, R-133, R-141.
Evidence grade: ANALYTIC, EXACT, CONDITIONAL, EXECUTED.
Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/a13_classii_innovation_compressed_common_feature_su2_covariance_signed_collar_band_boundary_verify.py
Expected output: R-142 integrated PASS; primary 50/50 and independent 46/46; deterministic note/PDF, manifest, exploration, authority, all-page rendering, and public-surface gates pass.
Falsification gate: any failed mixed-centering, shared-trace, Hessian, $SU(2)$ block, mass/covariance, scalar-score, band-sign, authority/hash, PDF, exploration, or public-surface assertion.

Tier before / after: T4 / T4; no promotion.

No-overclaim statement: no full production common-heat matrix, uniform
Loewner bound, source-null/low compatibility, full-owner counterexample,
positive graph gap, matching, anchor, A13 gate, OVERLAP_src, Nelson,
removal, interacting measure, Sector-A closure, or T5-T7 claim follows.

Next required action: assemble the non-Toeplitz $q=5,6,7$ four-channel
U/Phi cross-reveal matrix with anisotropic and returned-low blocks,
then compute its generalized source eigenpairs before extending to C10.